

ActInf Textbook Group ~ Cohort 1 ~ Meeting 20

(Chapter 8, part 2)

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hello everyone it's October 14 2022 we're in meeting 20. we're in our second discussion of Chapter eight we have a few topics and points in chapter eight and then maybe we can preview some chapter nine and have a lot of space for any other thoughts or questions people are having so we can jump to the questions and look at the um written chapter 8 questions or does anyone want to suggest a point for us to go to last time we know that some of the areas we wanted to pick up on involved learning and hierarchical modeling and hybrid modeling one question that came up I think here but maybe we can revisit is the Birdsong model is widely pointed at as an example of how communication is modeled in active inference or for active inference entities however it's quite literally singing from the same hymn sheet and the turn taking is based upon uh joint expectation of that song being sung that way and then the actions that the birds can take the kind of turn taking that they enact is against related to whether it's like well somebody has to say it is it going to be me or is it going to be that so how do you think we would broaden that scope of communication to account for situations where there isn't a pre-scripted song how do other Frameworks for communication cybernetic ish or otherwise deal with the semantics of conversation improvisation open-endedness in conversation uh actually I had a somewhat more general question about uh the bird songs and its relation to uh this particular systems that they've developed in the textbook uh you see some researchers believe that I'm sure you're much uh qualified to talk about uh this than me but there are lots of research going going around about the territorialization uh function of bird songs so if we take that as uh if we take a birth song as uh as its territorializing mechanism at least one of uh it's territorializing mechanism uh can we somehow put the dynamic I mean the dynamic boundary of the markup blanket uh I think would be affected uh from this point of view and then I'm not sure how that would translate into this specific model if we take that Dynamic nature of the mark of blanket boundary yeah I guess I would just build on that and say that um the communication serves a purpose and so um I think a model shouldn't

incorporate what that purpose is and how it's served by the communication and then whether it happens to be you know syncing up with some predefined pattern or or whatever is is a is a follow-on consideration the generative model in the singing from the same hymn sheet is like is very much like the sheet music example provided in the textbook there's a known sheet music script such that even when there's like a error in the playing of the music still the sheet music can be heard so just coarsely what generative models um I think it's pretty straightforward to see how a generative model could have the notation or the sheet music and then use the sheet music as its prior to kind of converge reality to what extent is communication that way is it totally disparate and differently structured or is it a little bit like a hybrid like you're doing inference on the sheet music that they're about to say and then you use your inference about what they're about to say to converge what they do say well you ask a good question which is well what are some alternative Frameworks and um I I know I I know I know some or should know some and I they're not coming to mind what comes to mind uh most immediately in this example is a phase lock loop in electronics which is simply a way of synchronizing one circuit's oscillation with another and um this is like a phase lock loop with a bigger vocabulary so figure out where we are in the in the pattern and then um synchronize with it and um but to the answer the question of what other Frameworks I mean there's this whole field of um conversation analysis um which um well it feels actually called CA conversation analysis and they have identified patterns of um how people communicate through a certain structure of I guess we call them rhetorical devices for achieving the purpose of the conversation so the purpose of the purpose of the communication is kind of Bedrock but um and there are multiple purposes some of which are instrumental some are which are you know social and then they have identified patterns for the way human conversation tends to proceed which are culturally influenced you know there's some cross-cultural aspects but then there's certain different cultures that have different takes on it so they identify things like okay now we're in the conversation opening phase now we're in a phase of turn where we're exchanging information and we have turn taking now we have Notions of um epistemic leadership or epistemic authority um you know who's who's the person who's asking who's the person who's answering or taking those uh those roles in the conversation and and there's certain rhetorical stances that are taken then we have so um that's a pretty well worked out um I I don't I guess even formalized system because they you know there's just there are um you know well identified patterns um it's it's not strictly worked out because it seems like every domain a conversation analyst goes to and every every group has their own kind of flavor of it

but it is kind of um uh it it does have this aspect of synchronizing with respect to the underlying known patterns of each of the parties in a conversation the bird song is also oh Ali please uh sorry adding to what Eric just mentioned uh that reminds me reminded me of the uh a much mechanism on a much much smaller scale namely the scale of cells and so on uh and uh particularly the model proposed by Stuart Kaufman because um he talks about the pluripotent nature of cells and how they communicate with each other uh in the context of phenomena of induction biological induction and uh I mean uh in the embryological development stage of the cells uh almost all cells are pluripotent and they're capable of becoming any of the different types of cells which characterize the adult individual so um that also the the uh from Kaufman's model perspective uh these kind of uh inductive signals or so to speak communicative signals between cells can act as a kind of non-specific stimulus which switches cell among a variety of internally available stable States uh so uh I mean a cell or let's put it this way the basic idea in this model is that the regulatory genes within a cell form a complex Network in which genes uh interacting by their products can turn one another on or off you see so uh yeah these patterns exhibit the kind of homeostatic stability associated with attractors of the systems so uh I think that translates perfectly into this scale as well this is small yeah the the mapping of communication as generalized synchrony may be compatible with a variety of communication mechanisms and so it abstracts us from the media of communication per se because we can talk about the informational synchronizations between for example the two internal states of the two entities or multilateral synchronizations with environmental States or so on like thinking about um the basic mechanics Bayesian physics the synchronizing metronomes they don't have a hem sheet they don't have sheet music their embodiments and coupling enables synchronization now one thing I might know you know you I think you asked a good question here which is you know how would you transfer translate something like um you know a human conversation into an active inference model so um but that that um kind of reminds me that the continuous valued um models because they're limited by these by what you can do in these differential equations they seem to be fairly simple whereas if I were to do something like thinking about human conversation in terms of active inference I would want more something more a very highly High dimensional or multi-state type of system where I can have more complex um patterns and they may not make these you know Loops these Volterra Loops or something called Volterra is that right or is this is Lawrence something Dynamics yeah yeah so it may not make those continuous Loops but I that you'd still get um you know good structured patterns out of discrete um systems so I've I'd

kind of be curious I think this chapter would be because if it what had if it had a um uh something about communication and synchrony pattern synchrony but in terms of discrete value um active inference models as a continuous valued ones yeah yeah very interesting like that's what I was wondering which is why is communication it is continuous time models are argued to be a natural fit for motor control our communication situations Naturally Fit for one or the other seems like I know you could imagine it either way and then also like um the synchronization means the internal state of one creature should come to resolve the uh resemble internal states of another A Primitive kind of theory of mind I think um not as a um uh fatal critique but these examples put the rabbit in the Hat a little bit too keenly like the planning example we explored wasn't truly planning it was like a parameterized um Subspace where single step optimization emulated planning now one could argue that they believe that that's actually like a better approach than than true explicit planning but that's not what was on the um cover and then similarly theory of mind though qualitatively may be a framing for communication we're talking still about the bird song where they have the same shared a priori generative model so this doesn't um in other words theory of mind is is um is axiomatically and implicitly embedded within this model rather than active inference being used as uh in a priori first principles to generate some realized computational theory of mind I feel like yeah yeah it's like um you know another aspect of this the The Continuous solute formulations is there's really not that many degrees of freedom but when you're communicating you know anything other than a fixed song there's a lot of degrees of freedom and um that's why we have symbol systems but simple systems is where you would use a discrete model framework for not a continuous value okay let's talk about hybrid models discrete and continuous models and learning and updating so these are two features chapter eight within a continuous time framework just like chapter seven layered in these kind of patterns or archetypes of discrete time models in chapter 8 analogously the most kernel continuous time formalism is introduced with movement control and then we're going to see some kind of extension extensions or patterns like design patterns that include continuous time this is the message passing okay so missing line here but we have here this top half is like the bottom of figure 4.3 or at the top I guess discrete time so we have B Matrix as a transition Matrix we have discrete time and we have um yeah it's discrete time t plus one T minus one but within each time point we see the continuous time you know e whatever we want to call this motif where the B transition Matrix and discrete time is functionally substituted with a derivative map such that rather than actually uh retro or forecasting State values through time this is much more like a Taylor series

approximation where extrapolation from the presence under the auspices of slower changing causes V which play a functional role of policies higher derivatives capture more and more variants further and further from the origin what is the ETA doing it's playing some functional role to G again allowing for a line here in the sense that it is selecting policy but it's not it's not policy that's being selected or discussed this way but these slower changing factors that influence how the derivatives are made but is you know a free energy variable let's just ask what do people think about this type of hybrid modeling um how does this align with the previous Decades of analog digital signals engineering and the interfaces the digitalization of analog noise and signals and um so on foreign well um I'll only offer one thought if it's all right one of the things that active inference let you do is look at a past trajectory and figure out how to interpret it in terms of your your variable your state variables so you can essentially you you look for the the probability distribution over variables that optimizes the free energy and that allows you to look back in time and reinterpret um evidence because you get something new and it you get you know in light of new evidence and then some of the earlier chapters they had this kind of example and they had the example of the of a dynamical a dynamic programming kind of a a model for this so I would expect that to be the case also in um systems and I guess I'm going back again to communication where uh to some extent you're able to reinterpret what you thought you heard before um you know we it's kind of rare in in language but it happens there are Garden Path sentences but usually you know you most communication systems try to be relatively hierarchical in that once you was once you interpret your continuous value signal then you don't leave a lot of ambiguity you kind of you know fix on that and then you have to deal with it later and sometimes you go back and do repair so I'm wondering if this figure 8.6 if that allows that kind of Dynamics where um you know you're kind of tracking something in the continuous value level you're tracking a signal why effectively by making predictions of what you're going to hear uh What You observe um and then can is that able in this framework to go and influence the symbol level the uh the B you know the s variables above and how far back in time is it able to do that nice that makes me think about phonemes down here and audio and then here are the um spellings so in a language or someone is speaking unclearly or you're not super familiar with the language then you really are like doing inference on the on the whole time series like it's like that kind of um retro casting which can include even revisionism and and delusions and false memories and all that now casting including the tension between representing things quote as they are versus like um having delusions in the present and then anticipation which of course hasn't even

happened unless you have a vision of time as you know something that always was or however it's really interesting that although this time T is an index we are doing inference across these and maybe it would speak to some um measure of information processing and memory and anticipation the extent to which like meaning is able to be the closer meaning is to a critical point like the more this graph structure if someone says but I was just kidding you would want that to apply to to be like oh I didn't know I was in a parenthesis and now that they were just kidding about that so it's actually like delineating those structural markers and then reinterpreting meaning perhaps moving towards the kind of linguistic modeling so yeah to come back to it so this could be a waveform the observables down here could be just a four-year analysis or or a pure auditory um waveform yeah I think phonemes is a that's what comes to mind also motor tasks so selecting a Target location and using motor Behavior functionally eyes to foveate the Target figure eight seven okay figure eight seven is a hybrid or mixed generative model higher level is this is kind of like the decision making motor active inference Fusion here we have a little bit of the italics and bold which we could dive into but don't need to do so at this moment so you know my recollection for last week was that that corresponded to um generative model versus process model you think that might map again here I know you didn't want to dive into it all right I believe it does let's just remind ourselves where it came up earlier in the chapter yeah it was the very beginning right here yeah with italics versus process in bold model in italics so eight seven so now here the um discrete model is on a time period of 200 milliseconds continuous model operates continuously um I think this speaks to how biological knowledge can be encoded in the model structure I mean a more comprehensive approach would have done parameter sweeps for what the optimal number of milliseconds is to do a discrete model on however one could say like well the physiology that we're interested in is accomplished with a gamma EEG cycle the gamma EEG cycle is this many milliseconds so we're just going to treat that as like a discrete Rhythm like one can imagine if the discrete time were one millisecond here you would be losing computational advantage and you'd be kind of like over slicing because if eye movement does happen on a time scale of hundreds of milliseconds discretizing to one millisecond is going to be too fine and discretizing to one second is not going to enable multiple actions to be taken within that frame so time scale and spatial scale of a system might be able to be encoded at the phase of its construction or structured learning more generally gaussian mixture model and the hierarchical yeah so before before we move on just about the isocades um I don't understand what is deemed by an active inference model for this I mean why not just say you got a

controller yeah it's a control Loop and that's good enough yes in um live stream number 50 which we're just preparing for it's like very classical cybernetic control Loop homeostatic Dynamics it's interesting to ask like again what makes the model Act of inference one could say that if there's a Markov blanket but we're you know between every one of these variables it is a model of action and perception like so many other models of action and perception so what is being shown here is it that the single imperative the free that energy-based imperatives just speaking narrowly here can mediate decision making in discrete continuous Hybrid models and that it's possible to have a categorical decision-making model in feedback with A continuous ocular motor model at the very least it places this type of model within the domain and grammar of active inference okay data clustering we may have touched upon also a hierarchical gaussian filter that I know some people have worked on okay let's look at some of the summary advances in continuous time models synthetic bird song we talked about this ocular motor delays conditioned reflex that's okay so this is in the chapter this is in the chapter I don't know about this exact paper perhaps but ocular Motors in the chapter and smooth eye Pursuits conditioned reflexes that is in this chapter the blank the conditioned and the unconditioned to stimuli building on the Songbird model but the Songbird model was in 2015. false inference from sub-optimal prior beliefs can but prior beliefs be sub-optimal um sorry uh actually McPherson has talked about bird songs uh before 2015 in uh one of his papers I think from 2013 or 12 uh yeah it's it's from 2013 which is co-authored with uh his son I guess Dominic aphrison so yeah I mean 2015 paper was not the first paper the name of the paper is the is the free energy formulation of Music generation and perception thank you thank you yeah wow yeah good um find a reminder interesting looks very much like the textbook you know opening quotes I can see how this is relates here researching too oh maybe even 2009 bird song chaotic attractors wow you know it's 10 10 plus years on the bird song paradigm okay so um deviations in perception more eye movements it's kind of fun like how we've talked about babbling and that across different domains like cicading no there's the ocular motor cicade but are there other kinds of cicadings [Music] for example I think that Knowledge Management niches facilitate semantic cicading yes enabled by visual cicading as well because you have to be like looking around on the page but it's just like what was chapter four okay okay that's what we're talking about chapter four what's notation I think it's just sort of like the analog to scanning around a little bit conversational analysis that's you know that's a very important topic is topic control and um or you know if you're uh if you look at um any you know most conversations have multiple threads going so it's like a Fugue in you know

Bach okay you know which voice is this are you you know every note is in a voice and the voice has got some topic and they intertwine in girdle escherbach Prelude Aunt fugue nice this is very um interesting I think it starts to plant some seeds towards again semantic cicading computational conversation analysis and generation adaptive knowledge environments like visual accessibility is one topic you know if somebody has um a visual difference then making sure that they can see it is the enabling system but let's just assume that we're within a space where we can talk about what is being seen so then um how does someone get a clear picture of what we're talking about that would be through some sort of cicade where you'd recognize um more uncertainty at the periphery and um like the fovea in this case is paying attention to the conversation and that's why like with a language we're unfamiliar with we get a little bit lost because like our circuit was too slow we don't have the fluency to cicade and so the video is moving too fast to track so we're just kind of like randomly listening and like trying to like compute but what if somebody were on a guided cicade journey through their semantic landscape I mean just to take this a little bit further um this is also maybe uh you know it's a critical element of uh cognitive architecture which is the ability to topic switch and you know in Psychology and cognitive science we there's this notion of executive control and one of the fundamentals of executive control is to be able to topic switch so you're on some task you get an interrupt you have to put that on a stack deal with your interrupt then when you're done and maybe the interrupt is a goal it could be a sub goal it could be some other topic that comes in something you have to deal with then when you're done you have to be able to pull that back push that back off the stack and get back to where you were so all that that context switching is really I mean there's measures of how people do this and there's debates about whether people can can multitask or not whether they're actually really multitasking or they're just multi-threading but they're there's they're dividing the amount of computational process by the number of processes you have going so you're actually not doing multiple things at once as effectively as if you just stuck with one so um yeah anyway and then it's like do we have that Intel hyper threading capacity is it a virtualized hyper thread on one thread are there actually different threads yeah I mean the conversation analysis other than us being right in the thick of it and it being highly engaging supported area let's just continue through this cases here action observation okay mirror neurons this may be very much related to the um bird song seeing from the same hymn sheet or whatever and then like you know doing the same dance okay attention this is a super fascinating area I think a lot of notebooks and and applications like just taking a trace of somebody's browsing on Wikipedia or across websites and then

reconstructing components of their generative model based upon their salience revealed choice is a huge area and then doing again that with the ocular motor with webcams uh and doing it with uh semantic landscapes hybrid modeling self-organization arguably with some pretty interesting advances in the last several months and days from Fields 11 others on the and four stage earlier but on the morphogenesis as Bayesian inference in our last few minutes here let's just look ahead to nine because I think this is setting us up well model-based data analysis or databased modeling or however we want to say this is about parameterizing structured models with observations so rather than just by Fiat dictate saying this is the structure of the model and it spits out observations like so and so we want to be able to actually run it backwards not as a generative model but as a recognition model so to speak in this expectation maximization framework and parameterize our generative model from sparse or dense observations and that's also called parametric empirical base because you're parameterizing the prior based upon the mean and variance and patterns of some empirical data okay the metabasian approach Bayes theorem this may even be something that could be introduced earlier even though it's also meta this is like the the map territory fallacy fallacy graphical abstract which is we're making a map here's Us in the bigger box even though there's a dashed box even Beyond us we're making a map of a cognitive entity in its map making of its generative process and that was the subject also of some very rich discussion several days ago with Jacob and Ali and Dalton Ollie anything to add on this uh no nothing much but uh one General comment I wanted to make about uh chapter nine is that uh you see because of my project that I've been working on for the past several months this chapter nine has been quite helpful in developing some of my ideas around how to uh how to exactly model my am I I mean the music emotional recognition system that I've been working on but I wanted to point out uh one small caveat here uh which is that uh you see at least in my opinion chapter nine or uh this so-called model based data analysis uh it's not a conducive to generate every I mean on every model that's been described throughout the throughout the book I mean it's just focused on some specific kinds of models so it might not be generalizable to every kind of model that we've seen throughout all the previous chapters good point some models might be just plug and play so to speak and other ones might take a little bit more work to get to use empirically okay let's just continue in our last minute so scan through so metabasian process we're drawing inferences about an inferential process it's a key point and that's the two levels of mapping it's like the map of the map variational LaPlace variational Bays factorized models for computational Simplicity and interpretability based upon the sparsity of coupling of variables

in a base graph and LaPlace using a parabolic approximator of the mode and a variance to ensure a simple optimizable heuristic not that it perfectly recapitulates the shape of the posterior which doesn't have to be inverted quadratic but the LaPlace approximation is smoothly optimizable for both the mode and the variance and here's where they mentioned that's the parametric empirical Bayesian approach and basically LaPlace approximation can be thought of as making a commitment to the quadratic form so choosing the family that your prior is going to be of and then bootstrapping your initial prior with your observed data bootstrapping not in the resampling sense but in the starting oneself out sense um this was explored more in other works but you have some likelihood function of all the parameters L of data optimizing action involves Computing beliefs about policy such that policies are proportionally taking up slices of the pie monotonically with respect to their free energy and then there's a Precision on action parameter where you can turn it all the way down to sort of like don't know don't care where even large differences in free energy between policies get equilibrated out that's like hot decision making and then absolute zero decision making would be a policy that's even 51 percent more likely would be always selected the variational free energy expected free energy it's abundant on surprise due to Jensen's inequality and evidence lower bound and all this other work in variational inference and then that pie slicing can be tuned up or down with a shaky hand soft Max temperature parameter if the Soft Max is one the pie is undistorted if the Soft Max is large you have a Relentless bias for better policies or it could be less than one here's some technical details of the LaPlace approximation parametric empirical base generalized linear modeling #spm here's recipe so this is kind of like chapter six but now it's model based data analysis collect behavioral data maybe the data's already been collected maybe there's some data sets that we already already can use formulate a pmdp I wonder if this is as chapter six not as chapter 7. it could be chapter seven as well if it's discrete time specify likelihood perhaps we can um in the coming weeks check out SPM mdp vbx and they mentioned the scripts data this is nice this will be fun chapter seven I mean it is the correct if they're using this type four chapter seven chapter four this looks pretty cool pretty interesting we could walk through some examples false inference computational pathology so it kind of ends like chapter eight with a table of references and a summary so it's a pretty short chapter has some formalisms but it's only from page 179 of the PDF to 196. so it's less than 20 pages and the next chapter I think is the longest chapter in the book which one is chapter 10 chapter 10 197. yeah that one's over 30 pages but it ends on a high note all right well conversations thanks for this we will come um back to chapter nine look through it and

then peep into 10. go through 10 and really solidify our understanding of the map and I guess indirectly of the territory and then for those who want to join will probably continue the discussion and um especially start looking towards those project ideas again like we talked just a little earlier about making a DOT zero a DOT zero sessions for the textbook so I I have to apologize I'm I'm traveling for a few weeks so I'm gonna miss the next um several sessions it's all good thank you Eric um any other comments people want to make or I'll close the recording thank you